



Inverse Problems and Data Assimilation

APMA DRP Fall 2026

Mentor Info



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Reach out anytime!



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Website



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Meeting Info



Mentees: 2 students



Day: TBD



Time: TBD



Location: 170 Hope Street

Overview

Models in applied mathematics often have input parameters that are uncertain, and observed data can be used to learn about these parameters and thereby improve predictive capability. The purpose of the DRP is to describe the mathematical and algorithmic principles of this area, with applications drawn from the physical, biological, and data sciences.

We will focus on the challenge of *inversion*, which is the process of extracting hidden information from noisy, indirect measurements. Unlike classical deterministic methods, we will adopt a Bayesian perspective where unknown parameters are treated as random variables. This approach allows us to not only find a best solution but also produce uncertainty quantification on the estimates.

Material

The primary reference that we will use throughout this DRP is Andrew Stuart's first book (SST), which is a theoretical introduction to the framework of Bayesian inverse problems and data assimilation.

Andrew Stuart's second book (SSBB) is a more machine-learning flavored treatment of the content, and offers a more computational treatment of how these theories scale to high-dimensional, data-driven applications.

Pre-Requisites

To gain the most from the DRP, fluency in probability (e.g., APMA 1650, 1660, 1690) and linear algebra (MATH 0520) is required. Some familiarity in analysis (e.g., MATH 1010, 1630, 1640) and coding (ideally Python) is recommended, but not required.

Expectations

Students are expected to engage deeply with the weekly reading assignments, which consists of one chapter of SST weekly, and should strive to grasp the underlying proofs and the concepts that support them.

The weekly meetings will be collaborative and informal. Most meetings will begin with students informally presenting the core results or proof strategies from the week's reading. There will time allocated afterwards to ask questions afterwards about the content or discuss research problems!

To get a handle of the practical side of the techniques, we will have a mini-project that is outlined later in the document that will involve a healthy mixture of theory and implementation.

Learning Objectives

In this course, we will study the following concepts:

- Formulation/well-posedness of probabilistic approaches to inverse problems and data assimilation
- Closed-form solutions for the posterior, filtering, and smoothing distributions in the linear Gaussian setting
- Optimization methods for point estimates and Gaussian approximations
- Sampling methods for uncertainty quantification

This subject is expanding so quickly that it is impossible to give an account of all developments. However, the selection of topics and structure of the course aim to provide the foundational tools necessary to understand and contribute to research in these areas, as well as stimulate interest in the subject.

Reading Schedule

The following is the weekly breakdown of the content covered in this DRP, with Week 0 being a review of the necessary linear algebra and probability background. While the lecture notes will be self-contained and have all the content needed, I encourage you to read some relevant papers in the field time permitting, and I am always happy to discuss!

Week 0	Review of Probability Theory & Linear Algebra
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Module 1: Bayesian Inverse Problems

Week 1	Well-Posedness of Bayesian Inverse Problems	SST Ch. 1
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Week 2	The Linear-Gaussian Setting	SST Ch. 2
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Week 3	Optimization Perspective	SST Ch. 3
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Week 4	Gaussian Approximation	SST Ch. 4
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Week 5	Monte Carlo/Importance Sampling	SST Ch. 5
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Week 6	Markov Chain Monte Carlo Methods	SST Ch. 6
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Module 2: Data Assimilation

Week 7	Well-Posedness of Smoothing/Filtering Problems	SST Ch. 7
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Week 8	The Linear-Gaussian Setting	SST Ch. 8
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Week 9	Optimization Perspective	SST Ch. 9
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Week 10	Extended/Ensemble Kalman Filters	SST Ch. 10
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Week 11	Particle Filter	SST Ch. 11
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Week 12	Optimal Particle Filter	SST Ch. 12
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Module 3: Wrap Up

Week 13	Final Presentations
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